LIMKOKWING UNIVERSITY OF CREATIVE TECHNOLOGY
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FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

PROG101: PRINCIPLES OF PROGRAMMING LOGIC & DESIGN

Title : Assignment – Design Logical Solutions for Sierra Leone Community Challenges: Child Fever Referral and Follow-up System for Suspected Malaria
Issue Date : Week
Due Date : Week
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Class : BBIT1101F
Semester/Year : 1/1

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Abstract

Malaria remains a major public health burden in Sierra Leone, especially for children under five. Fever in a young child requires prompt assessment, yet distance, limited communication, and incomplete records can complicate referral and follow-up. This assignment proposes a Child Fever Referral and Follow-up System to help community health workers document fever reports, identify danger signs using locally approved guidance, direct urgent cases to immediate care, and track whether children reach a trained provider. The system is a low-resource referral and tracking design, not a diagnostic or treatment tool. It does not confirm malaria or recommend medicines; diagnosis, testing, and treatment remain with trained health providers. At intake, the worker records age, a non-identifying case ID, community, report time, and danger-sign response. The case ID links later follow-up to the saved intake record, avoiding repeated entry of community and time. Follow-up records whether care was reached and assigns a PENDING or CLOSED status. Weekly summaries group totals by community and reporting period. The logical design demonstrates input validation, Boolean checks, selection, repetition, and modular functions. Three Flowgorithm-style charts show intake and screening, case follow-up, and weekly reporting. The workflow can be used with paper or an offline-capable device where internet access or electricity is unreliable. To protect children and caregivers, records should contain only necessary details and remain accessible only to authorized workers. The proposed system supports Sustainable Development Goal 3, particularly target 3.2, which focuses on preventing avoidable deaths among children under five, and target 3.3, which includes combating malaria and other communicable diseases. Its expected benefit is clearer referral coordination and better visibility of unconfirmed follow-up, not a guaranteed reduction in illness or deaths. Real use would require local clinical approval, health-worker training, privacy safeguards, and coordination with health facilities.

1. Rationale

a) Problem description

The selected community problem is delayed assessment and referral for fever in children under five where malaria is suspected. A child with fever may have malaria or another illness. Caregivers may face distance, transport, cost, or information barriers, while community health workers may have limited means to track whether referrals are completed. The assignment therefore focuses on the information and coordination gap between a first report of fever and assessment by a trained provider. It does not assume every fever is malaria or that a digital system alone can resolve structural barriers.

WHO country data identify malaria as Sierra Leone's leading estimated cause of death in 2023. WHO's 2024 malaria profile reports approximately 2.1 million confirmed cases and 2,637 reported deaths. WHO also describes rural malaria prevalence as substantially higher than urban prevalence and notes the disease burden among children under five. These sources use different indicators and reference periods; they show why malaria is a defensible high-priority topic, but do not establish a universal ranking of all health problems by one measure.

b) Background

Malaria is preventable and treatable, but suspected cases need prompt assessment. WHO guidance emphasizes diagnostic testing for suspected malaria and explains that fever can arise from illnesses other than malaria. Severe illness can progress quickly, particularly in young children. For this reason, the proposed workflow sends danger-sign cases for immediate clinical attention and directs other eligible children for prompt assessment by a trained provider. It leaves diagnosis, testing, and treatment to authorized health personnel using current Sierra Leone protocols.

The burden is not only medical. A caregiver may need to decide whether symptoms are urgent, locate a health facility, and arrange transport. In rural settings, these steps may take time. A paper register or offline-capable phone can help a community health worker record the initial report, provide a consistent referral message, and identify cases whose outcomes remain unknown. The system cannot create a health facility, guarantee transport, or replace clinical staff; it can make the handoff more visible and organized.

c) Community impact

Children under five are the primary beneficiaries. Caregivers may benefit from clearer referral instructions and a contact point for follow-up. Community health workers can use a structured

case list to distinguish urgent cases from cases needing prompt assessment and can identify when a referral outcome has not been confirmed. Health-facility staff and district program teams may use de-identified summaries to understand referral completion patterns and improve coordination. The system should be designed so that these operational summaries do not expose children's identities to people who do not need them.

The intended community impact is improved referral visibility, earlier escalation of danger-sign cases, and fewer cases lost between community reporting and facility assessment. These are expected contributions, not guaranteed health outcomes. Evaluation should first measure whether records are complete, urgent referrals are escalated promptly, and follow-up lists are used. Any later claim that the system reduced illness or deaths would require suitable health data and evaluation beyond this logical design.

d) Proposed solution

The proposed Child Fever Referral and Follow-up System is a simple register and workflow for community health workers. It can operate on paper where devices, electricity, or connectivity are unavailable, with an optional offline digital form that synchronizes only when access is available. Each report receives a non-identifying case code. The worker records age, community, report time, danger-sign answers, referral destination, and referral status. The system displays an urgent referral message when a danger sign is present; otherwise, it prompts assessment by a trained provider. A trained provider performs assessment and testing according to national guidance.

A follow-up list shows referrals that have not been confirmed. The system does not send unapproved clinical advice, prescribe medicine, or label a child as having malaria. Only authorized health workers can access identifiable case details. Reports for planning use aggregated counts. Local health authorities should approve the danger-sign checklist, referral destinations, data-retention period, and follow-up timelines before deployment.

e) Computational thinking process

• Decomposition: separate the work into case intake, age validation, danger-sign screening, referral, follow-up, and summary reporting.

• Pattern recognition: review repeated missed referrals or recurring case counts by community and reporting period.

• Abstraction: collect the minimum information needed for safe referral tracking; omit unrelated personal details.

• Algorithm design: apply a consistent sequence of decisions, while ensuring danger signs lead directly to urgent referral.

f) Algorithm design

The algorithm accepts a fever report for a child under five, validates the age group, checks danger signs from a locally approved list, and selects an urgent or prompt-assessment pathway. It records the referral destination and loops through pending cases for follow-up. A case closes when an authorized worker records an outcome. The workflow uses sequence, selection (IF/ELSE), repetition (FOR EACH pending case), and modular functions. The danger-sign rules must be reviewed against current Ministry of Health guidance before use.

g) SDG alignment

The project supports Sustainable Development Goal 3, Good Health and Well-being. Target 3.2 calls for ending preventable deaths of newborns and children under five; Target 3.3 specifically includes combating malaria. The system contributes by supporting referral coordination and follow-up for a vulnerable age group. It is an enabling information workflow, not a substitute for prevention, diagnosis, treatment, vaccines, insecticide-treated nets, or a functioning health service.

h) Open-source justification

Open-source publication could allow Sierra Leonean students, local developers, health workers, and civic organizations to inspect, adapt, translate, and improve the workflow. An approved licence, clear documentation, change history, and community contribution process would make reuse easier and reduce dependence on one vendor. Contributions that change clinical content should require review by qualified local health authorities before release. Open source improves transparency and reuse; it does not by itself provide funding, maintenance, clinical validation, or privacy compliance.

i) Digital Public Goods justification

The system could be designed toward Digital Public Good principles by aligning with SDGs, using an approved open licence, documenting its operation, supporting platform independence and data export, and protecting privacy and security. The Digital Public Goods Alliance evaluates solutions against its Standard. This assignment describes a path toward those principles and does not claim that the proposed system is already recognized or listed as a DPG. Long-term sustainability would depend on local ownership, training, reliable referral pathways, maintenance, and a clear data-governance plan. The system should work without continuous

internet access, use simple devices or paper forms, and keep the paper process available as a fallback. Community feedback and periodic review by health authorities would help keep instructions understandable and clinically current.

2. Algorithm

1. Start a new fever case and create a unique case ID.
2. Record age, community, report date and time, and fever report.
3. If age is five years or older, direct the case to the appropriate local health workflow and end this under-five process.
4. For an eligible child, record danger-sign responses using the health-authority-approved checklist.
5. If one or more danger signs are present, mark the case URGENT and direct immediate referral. Do not delay referral to complete nonessential fields.
6. If no danger sign is reported, mark PROMPT ASSESSMENT and arrange assessment by a trained provider under national guidance.
7. Record referral destination and outcome status.
8. For every pending referral, check whether care was reached, if not confirmed, keep it on the follow-up list under local procedures.
9. When an authorized worker records the provider outcome, close the case.
10. At the end of the reporting period, create an aggregated summary for authorized staff.

Safety boundary: the workflow supports referral tracking. It does not diagnose malaria, interpret a test, select medicine, or delay urgent care. Clinical actions and follow-up timing must follow current national protocols.

3. Pseudocode

```
CONSTANT UNDER_FIVE_LIMIT = 5

FUNCTION RegisterFeverCase(ageYears, community, reportTime, dangerSigns)
   caseID = CreateUniqueCaseID()
   SaveMinimumCaseDetails(caseID, ageYears, community, reportTime)

   IF ageYears >= UNDER_FIVE_LIMIT THEN
      RETURN "Use the appropriate health referral process"
   ENDIF

   IF AnyDangerSign(dangerSigns) = TRUE THEN
      SetStatus(caseID, "URGENT REFERRAL")
      Display("Refer immediately to an appropriate health facility")
```

```
        Display("Do not delay referral to finish nonessential data entry")
    ELSE
        SetStatus(caseID, "PROMPT ASSESSMENT")
        Display("Arrange prompt assessment by a trained provider")
        Display("Follow current national testing and care guidance")
    ENDIF

    SaveReferralDestination(caseID, GetReferralDestination())
    reachedCare = AskIfCareWasReached(caseID)

    IF reachedCare = FALSE THEN
        AddToFollowUpList(caseID)
    ELSE
        SaveProviderOutcome(caseID, RecordProviderOutcome())
        SetStatus(caseID, "CLOSED")
    ENDIF

    RETURN CaseSummary(caseID)
END FUNCTION

FUNCTION ReviewPendingCases(pendingCases)
    FOR EACH case IN pendingCases
        IF ReferralNotConfirmed(case) THEN
            DisplayForAuthorizedFollowUp(case.caseID)
        ENDIF
    ENDFOR
END FUNCTION

FUNCTION CreateWeeklySummary(cases)
    reports = Count(cases)
    urgent = CountWhereStatus(cases, "URGENT REFERRAL")
    completed = CountConfirmedReferrals(cases)
    overdue = ListCasesNeedingFollowUp(cases)
    RETURN DeidentifiedSummary(reports, urgent, completed, overdue)
END FUNCTION
```

4. Flowcharts

The charts show separate stages of the same proposed system. The first handles intake and danger-sign screening. The second tracks whether a referral was completed. The third prepares a de-identified weekly summary. All clinical decisions remain with trained providers and local guidance.

Flowchart 1: Child fever intake and danger sign screening

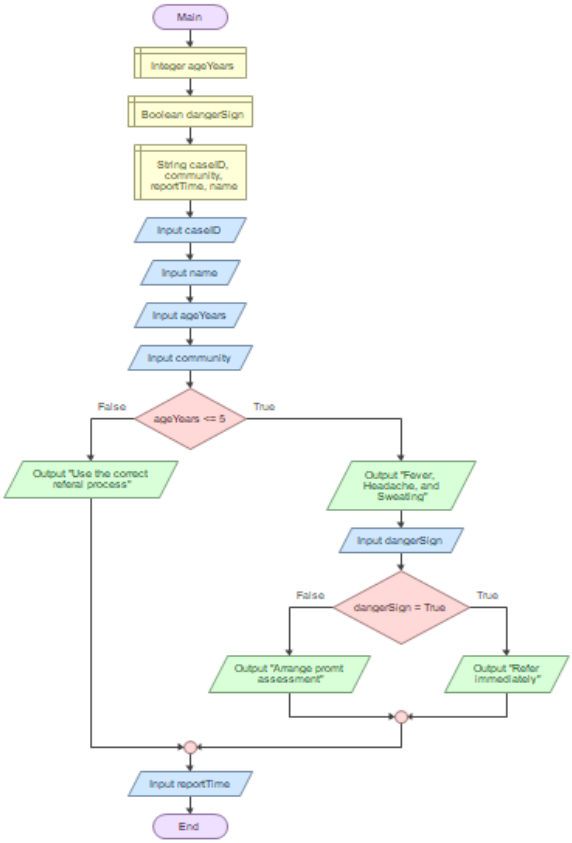


Figure 1. Intake and danger-sign screening.

Flowchart 2: Referral tracking and follow-up

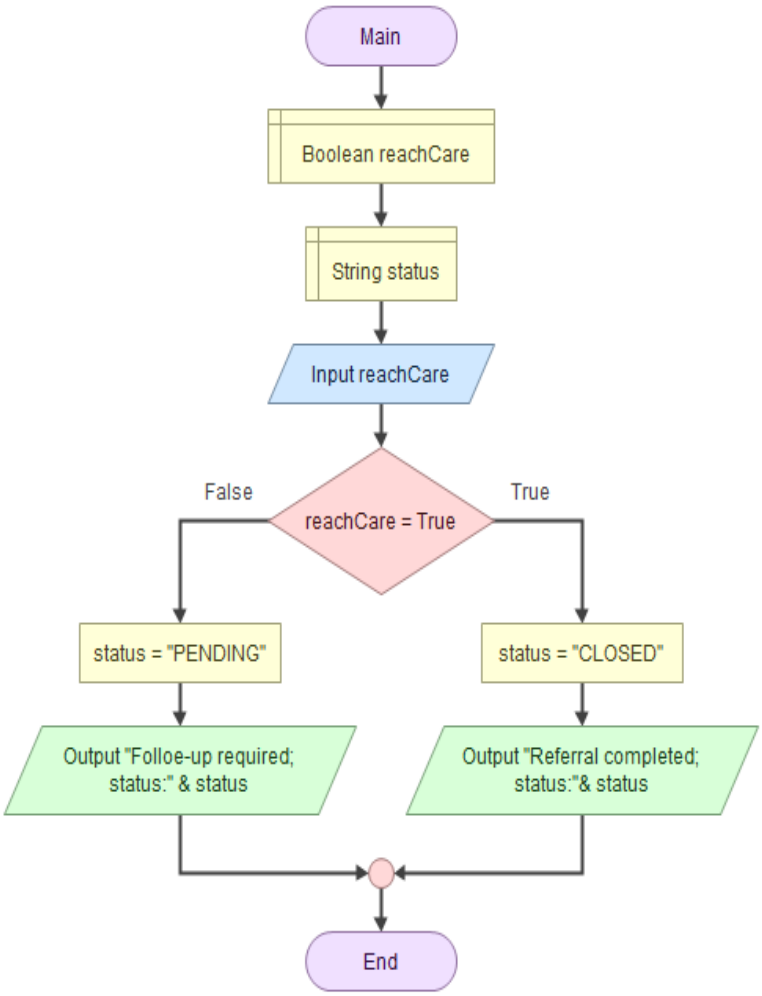


Figure 2. Referral tracking and follow-up.

Flowchart 3: Weekly Referral summary for health workers

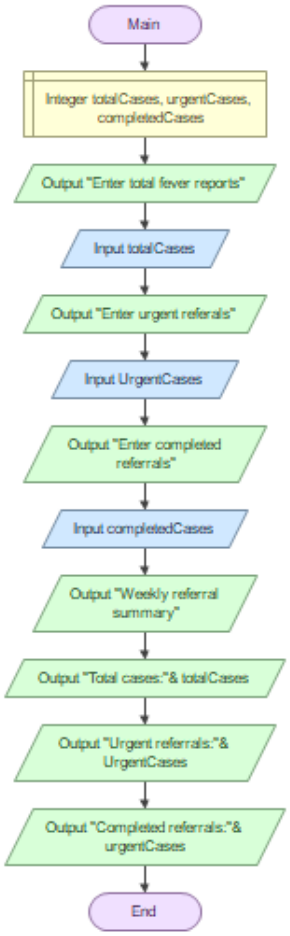


Figure 3. Weekly summary reporting.

5. Functional modules

Module or function	Purpose
CreateUniqueCaseID()	Generates a reference code that avoids placing a child's name in routine summary reports.
SaveMinimumCaseDetails(...)	Stores only the fields approved for referral and follow-up.
AnyDangerSign(signs)	Returns TRUE if any approved danger sign is recorded.
SetStatus(caseID, status)	Updates the case to urgent referral, prompt assessment, pending follow-up, or closed.
GetReferralDestination()	Records the selected health facility or service.
AskIfCareWasReached(caseID)	Records whether referral completion has been confirmed.
AddToFollowUpList(caseID)	Flags an unconfirmed referral for authorized follow-up.
SaveProviderOutcome(caseID, outcome)	Stores the outcome supplied by an authorized provider or worker.
CreateWeeklySummary(cases)	Counts cases and referral outcomes and produces a de-identified operational summary.

Data items and types

Data item	Suggested type	Example or rule
ageYears	Integer	0 to 4 for this workflow
community	Text	Community name/code, following local data rules
reportTime	Date/time	Time the report was received
dangerSigns	Boolean list	Yes/No responses from approved checklist
caseID	Text	Unique non-identifying reference
status	Text/constant	URGENT, PROMPT ASSESSMENT, PENDING, CLOSED
reachedCare	Boolean	True when referral is confirmed

6. Expected outputs

Sample scenario	Expected output
A 2-year-old with fever cannot drink.	URGENT REFERRAL. Direct the caregiver to immediate assessment at an appropriate

	health facility; do not delay for data entry.
A 3-year-old has fever, and no sign of danger is reported.	PROMPT ASSESSMENT. Arrange assessment by a trained provider following national guidance.
A caregiver cannot confirm the child reached care.	Case remains pending and appears on the authorized follow-up list.
A provider outcome is recorded.	Case outcome is saved and the case status changes to CLOSED.
A 6-year-old is entered.	The system directs the user to the appropriate health referral workflow; the under-five process ends.

Access, privacy, and operational assumptions

- Only authorized health workers can view identifiable case details.
- Use a case code in routine reports; produce counts rather than named case lists for planning.
- Allow paper/offline use and later entry or synchronization when connectivity is available.
- The local health authority must approve the danger-sign checklist, referral list, follow-up timing, retention period, and access roles.
- The system is not a diagnosis tool and must not delay urgent referral.

8. References

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5 World Health Organization. Universal access to malaria diagnostic testing: an operational manual. https://www.who.int/docs/default-source/documents/publications/gmp/universal-access-to-malaria-diagnostic-testing.pdf

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7 Digital Public Goods Alliance. Digital Public Goods Standard. https://www.digitalpublicgoods.net/standard

8 Limkokwing University of Creative Technology, Sierra Leone. PROG101 Principles of Programming Logic and Design assignment brief, September 2026.